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UID EXP-10

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AIM

To design and implement interactive data visualizations (bar chart and doughnut chart) for an inventory management system using JavaScript.

•**PROCEDURE**

•**1. HTML Setup**

Create an HTML file that defines the basic structure of the webpage. Add `<canvas>` elements where the charts will be displayed.

•**2. Include Libraries**

Integrate required libraries such as Chart.js into the HTML file using a CDN link to enable chart rendering.

•**3. Data Preparation**

Create a JavaScript file (`script.js`) and define inventory-related data such as product names, stock quantities, and product categories.

•**4. Bar Chart Creation**

Use JavaScript and Chart.js to build a bar chart that visually represents the stock levels of different products.

•**5. Doughnut Chart Creation**

Generate a doughnut chart to show how inventory is distributed among various categories.

•**6. Execution**

Link the JavaScript file to the HTML file and open the webpage in a browser to view the charts.

index.js | index.html | app.js

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Inventory UX Experiment</title>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>
        body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #f4f7f6; padding: 20px; }
        .dashboard-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(400px, 1fr)); gap: 20px; }
        .chart-container { background: white; padding: 20px; border-radius: 12px; box-shadow: 0 4px 6px rgba(0,0,0,0.05); }
        h2 { color: #333; font-size: 1.2rem; margin-bottom: 20px; }
    </style>
</head>
<body>

<div class="dashboard-grid">
    <div class="chart-container">
        <h2>Stock Levels by Product</h2>
        <canvas id="stockBarChart"></canvas>
    </div>

    <div class="chart-container">
        <h2>Inventory by Category</h2>
        <canvas id="categoryDoughnutChart"></canvas>
    </div>
</div>

<script src="app.js"></script>
</body>
</html>
```

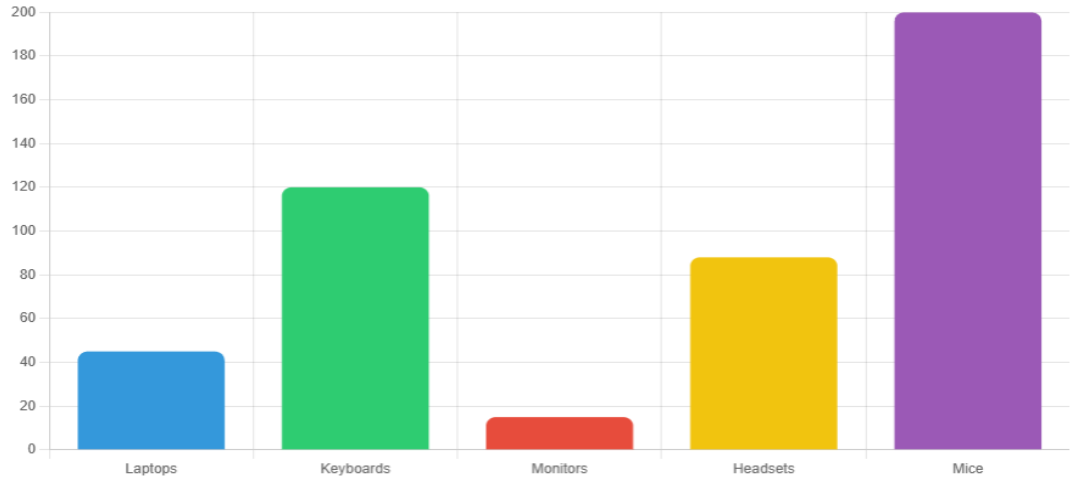
index.js | index.html | app.js

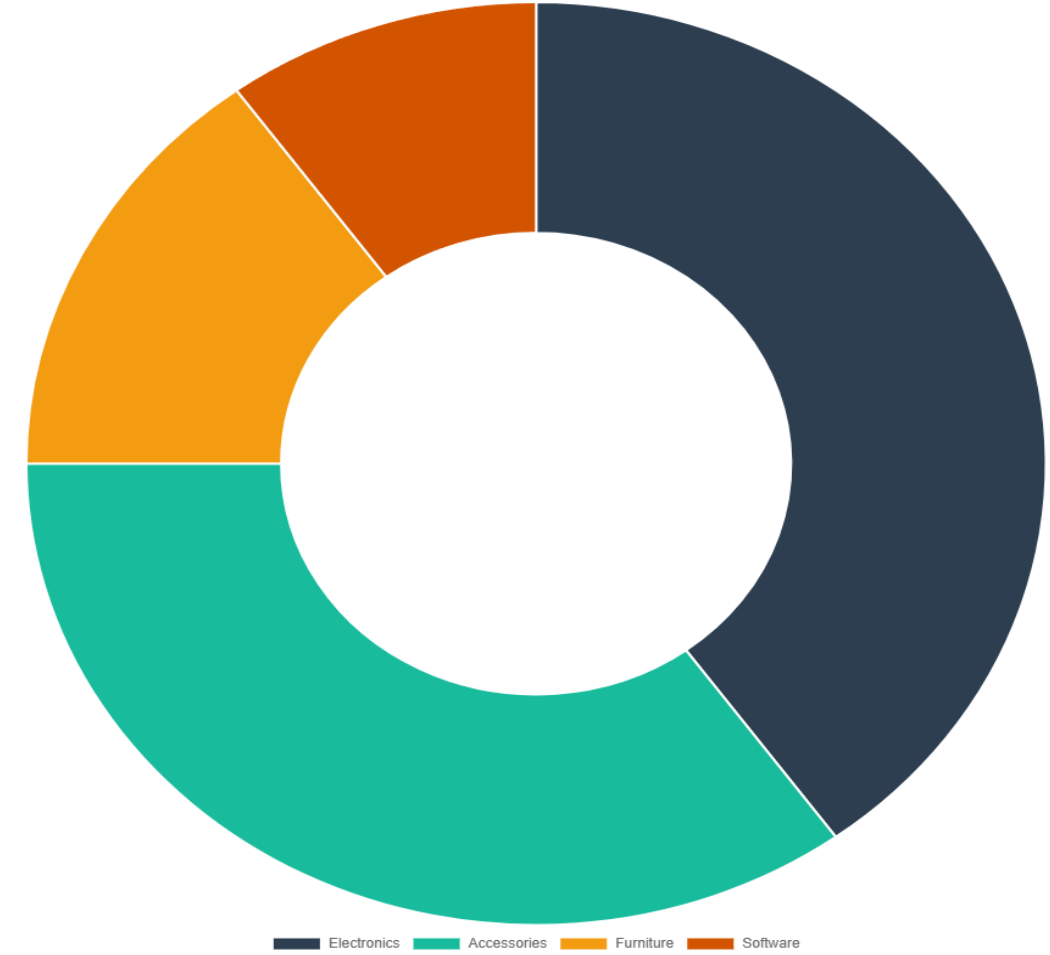
```javascript
const stockData : {datasets: [{label: string, data: number[..., labels: ...} = {
    labels: ['Laptops', 'Keyboards', 'Monitors', 'Headsets', 'Mice'],
    datasets: [{
        label: 'Current Stock Count',
        data: [45, 120, 15, 88, 200],
        backgroundColor: [
            '#3498db', // Blue
            '#2ecc71', // Green
            '#e74c3c', // Red (Low stock alert style)
            '#f1c40f', // Yellow
            '#9b59b6'  // Purple
        ],
        borderRadius: 8
    }]
};
const categoryData : {datasets: [{data: number[], backgroundCo..., labels: ...} = {
    labels: ['Electronics', 'Accessories', 'Furniture', 'Software'],
    datasets: [{
        data: [40, 35, 15, 10],
        backgroundColor: ['#2c3e50', '#18bc9c', '#f39c12', '#d35400'],
        hoverOffset: 10
    }]
};
window.onload = function() : void {
    const ctxBar = document.getElementById( elementId: 'stockBarChart').getContext( contextId: '2d');
    new Chart(ctxBar, {
        type: 'bar',
        data: stockData,
        options: {
            responsive: true,
            plugins: { legend: { display: false } },
            scales: { y: { beginAtZero: true } }
        }
    });
    const ctxDoughnut = document.getElementById( elementId: 'categoryDoughnutChart').getContext( contextId: '2d');
    new Chart(ctxDoughnut, {
        type: 'doughnut',
        data: categoryData,
        options: {
            responsive: true,
            plugins: {
```

index.js | index.html | app.js

```javascript
const stockData : {datasets: [{label: string, data: number[..., labels: ...} = {
    datasets: [{
        borderRadius: 8
    }]
};
const categoryData : {datasets: [{data: number[], backgroundCo..., labels: ...} = {
    labels: ['Electronics', 'Accessories', 'Furniture', 'Software'],
    datasets: [{
        data: [40, 35, 15, 10],
        backgroundColor: ['#2c3e50', '#18bc9c', '#f39c12', '#d35400'],
        hoverOffset: 10
    }]
};
window.onload = function() : void {
    const ctxBar = document.getElementById( elementId: 'stockBarChart').getContext( contextId: '2d');
    new Chart(ctxBar, {
        type: 'bar',
        data: stockData,
        options: {
            responsive: true,
            plugins: { legend: { display: false } },
            scales: { y: { beginAtZero: true } }
        }
    });
    const ctxDoughnut = document.getElementById( elementId: 'categoryDoughnutChart').getContext( contextId: '2d');
    new Chart(ctxDoughnut, {
        type: 'doughnut',
        data: categoryData,
        options: {
            responsive: true,
            plugins: {
                legend: { position: 'bottom' }
            }
        }
    });
};
```

OUTPUT:


Stock Levels by Product
200
180
160
140
120
100
80
60
40
20
0
Laptops
Keyboards
Monitors
Headsets
Mice


Inventory by Category
Electronics
Accessories
Furniture
Software